Zephyr 101

Docker Setup

Install docker

Download and install Docker client for your operating systems. There are plenty tutorials on YouTube on how to do it.

Get the one for your OS from here: https://docs.docker.com/desktop/

Create a Container

Use the following command to fetch the image we created and create a container on top of it. This container is where you will work.

```
docker run -it \
           -v ~/Downloads/workspace:/home/pyjamabrah/workspace \
           --name lab pyjamabrah/sandbox:latest
```

You can work in the docker container through the terminal. When done use `exit` to exit out of the container. This will also **stop** the container.

Reopening the Container

In this case, `lab` is the name of the container. To work with it again, you will need to start it and then attach a terminal to it. Use these commands -

- To start the container...

```
docker start lab
```

- to attach a terminal to the container

```
docker attach lab
```

Zephyr

Install Utilities

```
sudo apt update

sudo apt install -y git \
                    cmake \
                    ninja-build \
                    ccache \
                    dfu-util \
                    device-tree-compiler \
                    python3 \
                    python3-pip \
                    python3-setuptools \
                    python3-wheel \
                    python3-venv \
                    python3-requests \
                    python3-pyelftools \
                    wget \
                    file \
                    qemu-system-arm tree
```

Get the Zephyr SDK

Documentation on ZDK: https://docs.zephyrproject.org/latest/develop/toolchains/zephyr_sdk.html#supported-architectures

Following commands are borrowed from the the official SDK documentation.

```
cd /home/pyjamabrah/workspace/
wget https://github.com/zephyrproject-rtos/sdk-ng/releases/download/v0.17.4/zephyr-sdk-0.17.4_linux-`uname -m`.tar.xz
```

Extract

```
tar xvf <downloaded file>
```

Setup the SDK

Get inside the directory with extracted content and execute the setup script.

```
./setup.sh
```

When asked to confirm, use `y`.

Virtual Environment

Create a folder for the virtual environment in the `/home/pyjamabrah/workspace/` path

```
mkdir ~/zephyr_env
cd ~/zephyr_env
```

Set up python `venv` so all the python packages can be installed locally

```
python3 -m venv .venv
source .venv/bin/activate
```

Install West inside the `venv`

```
pip3 install west pyelftools jsonschema
```

Activate

activate the `venv` by executing the following command

```
source ~/zephyr_env/.venv/bin/activate
```

Fetch Zephyr OS Source

From within `/home/pyjamabrah/workspace/zephyr_env`, execute the following command to clone the Zephyr

```
git clone https://github.com/zephyrproject-rtos/zephyr.git
```

Tell the environment where everything lives

```
# 5.1 Export the Zephyr base directory
export ZEPHYR_BASE=$(pwd)/zephyr

# 5.2 Point to the SDK (so West knows where the toolchain is)
export ZEPHYR_SDK_INSTALL_DIR=/opt/zephyr-sdk

# 5.3 Make the compiler tools available on PATH
export PATH="$ZEPHYR_SDK_INSTALL_DIR/bin:$PATH"
```

Initialize West and fetch sub-modules

```
cd $ZEPHYR_BASE

# 6.1 Initialise West with the local Zephyr tree
west init -l .

# 6.2 Pull all sub-modules (core Zephyr repo, samples, bindings, etc.)
west update

# 6.3 Export Zephyr bindings to the workspace
west zephyr-export
```

Build & Run

- Build

```
west build -b qemu_cortex_m3 samples/hello_world --pristine
```

- Run

```
west build -t run
```

- Expect an output like so

```
-- west build: running target run
[0/1] To exit from QEMU enter: 'CTRL+a, x'[QEMU] CPU: cortex-m3
qemu-system-arm: warning: nic stellaris_enet.0 has no peer
*** Booting Zephyr OS build v4.2.0-4122-g9354e0642203 ***
Hello World! qemu_cortex_m3/ti_lm3s6965
```

> **Note**
>
> To exit QEMU use the following Key combination to exit QEMU - `ctrl+a`, `x`